

ONLINE COMPLAINT REGISTRATION AND MANAGEMENT SYSTEM (OCRMS)

PROJECT DOCUMENTATION

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CHAPTER 1

INTRODUCTION

1.1 Project Title

OnlineComplaintRegistration and Management System (OCRMS)

1.2 Project Description

TheOnlineComplaint Registration and Management System (OCRMS) is a web-based platform developed to streamline the process of complaint registration, management, tracking, and resolution. The system provides a centralized environment where users can submit complaints, track their progress, communicate with assigned support personnel, and provide feedback after resolution.

Traditional complaint management systems often suffer from delays, lack of transparency, and inefficient communication between users and support teams. OCRMS addresses these challenges by introducing automated complaint assignment, real-time communication, complaint tracking, and intelligent support assistance.

The platform is designed using modern web technologies and follows a scalable client-serverarchitecturetoensure reliability, security, and responsiveness.

1.3 Objectives

The main objectives of OCRMS are:

- **To provide a user-friendly platform for registering complaints.**
- **To automate complaint assignment to available specialists.**
- **To improve transparency in complaint resolution.**
- **To enable real-time communication between users and support agents.**
- **To provide users with complaint status tracking capabilities.**
- **To collect customer feedback for service improvement.**
- **To reduce complaint resolution time through workload-balanced routing.**

CHAPTER 2

PROJECT OVERVIEW

2.1 Purpose

OCRMS aims to digitize and automate the complaint handling process. The system allows

organizations to efficiently manage complaints while providing customers with visibility into every stage of the complaint lifecycle.

The application eliminates manual complaint allocation and introduces intelligent routing mechanisms to distribute workloads evenly among support personnel.

2.2 Key Features

- **User Management**
- **User Registration**
- **Secure Login System**
- **Google OAuth Authentication**
- **Role-Based Access Control**

Complaint Management

Complaint Registration

Category-Based Complaint Submission Automatic

Specialist Assignment Complaint Tracking System

Status Updates

-

Communication System

- **Real-Time Chat using**
- **Socket.io Agent-Customer**
- **Communication Instant**
- **Message Updates**

Smart Assistance

- **Integrated Chatbot**
- **FAQ Assistance**
- **Complaint Filing Guidance**
- **Password Reset Guidance**

Feedback System

- **Complaint Resolution Reviews**
- **Five-Star Rating System Customer**
- **Feedback Collection**

Modern User Interface

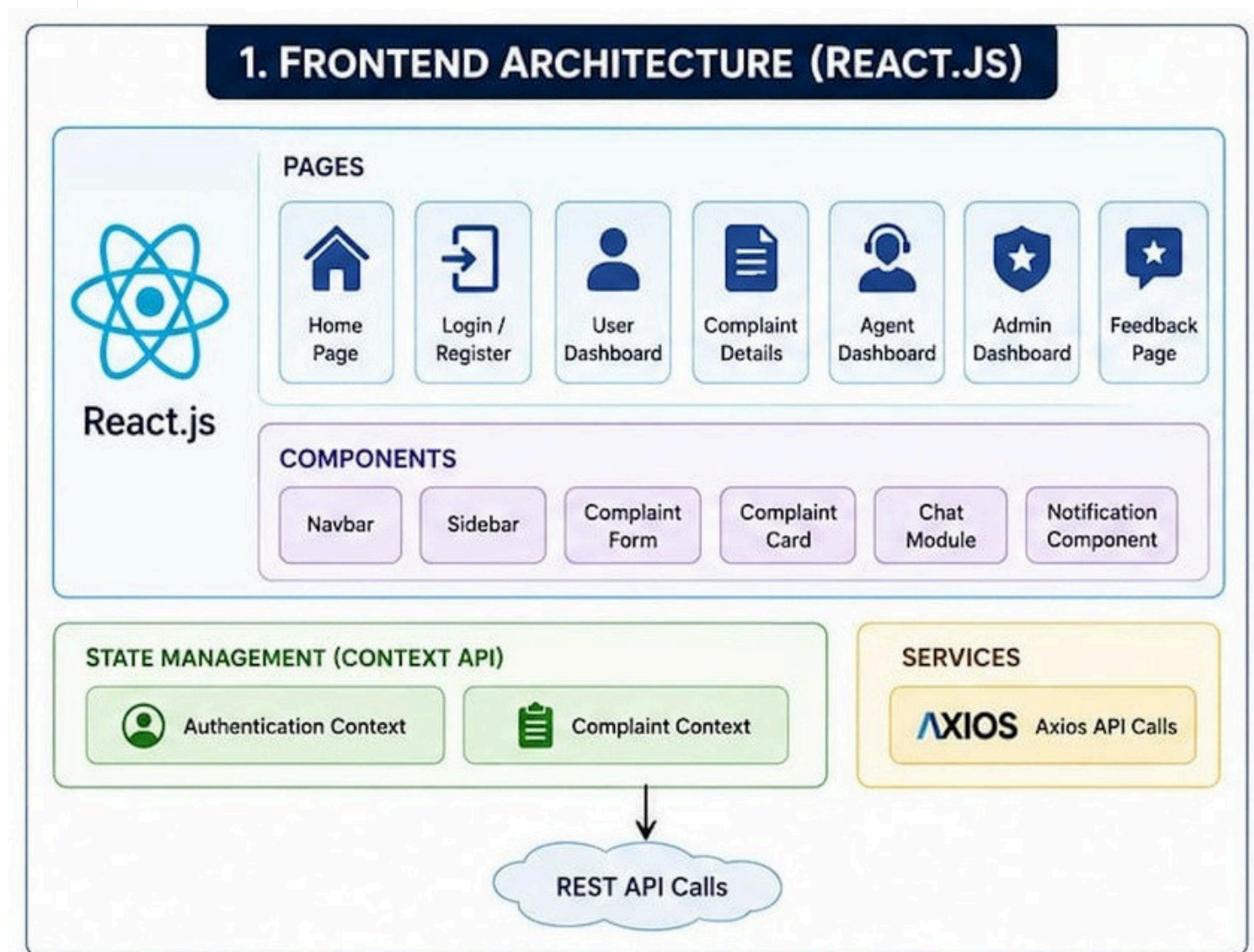
- Responsive Design
- Glassmorphism Effects
- Animated Components
- Mobile-Friendly Interface

CHAPTER 3

SYSTEM ARCHITECTURE

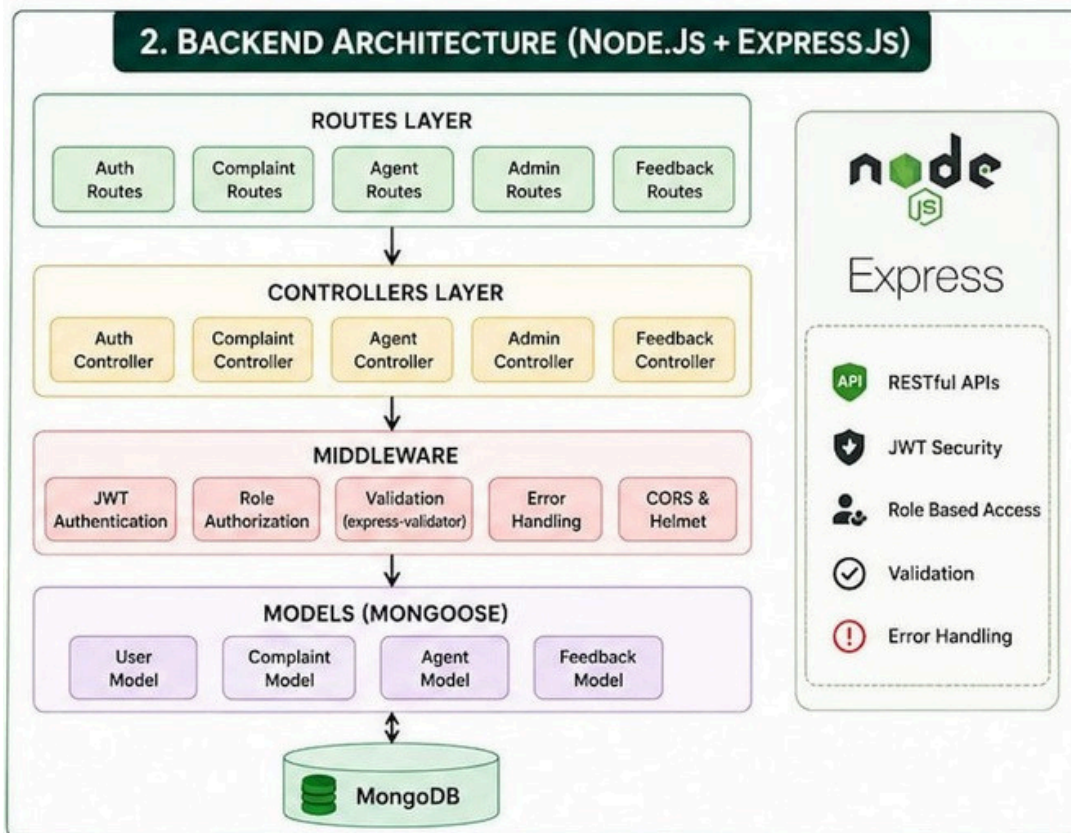
3.1 Frontend Architecture

The frontend is developed using React.js and Vite, providing a fast and responsive user experience.



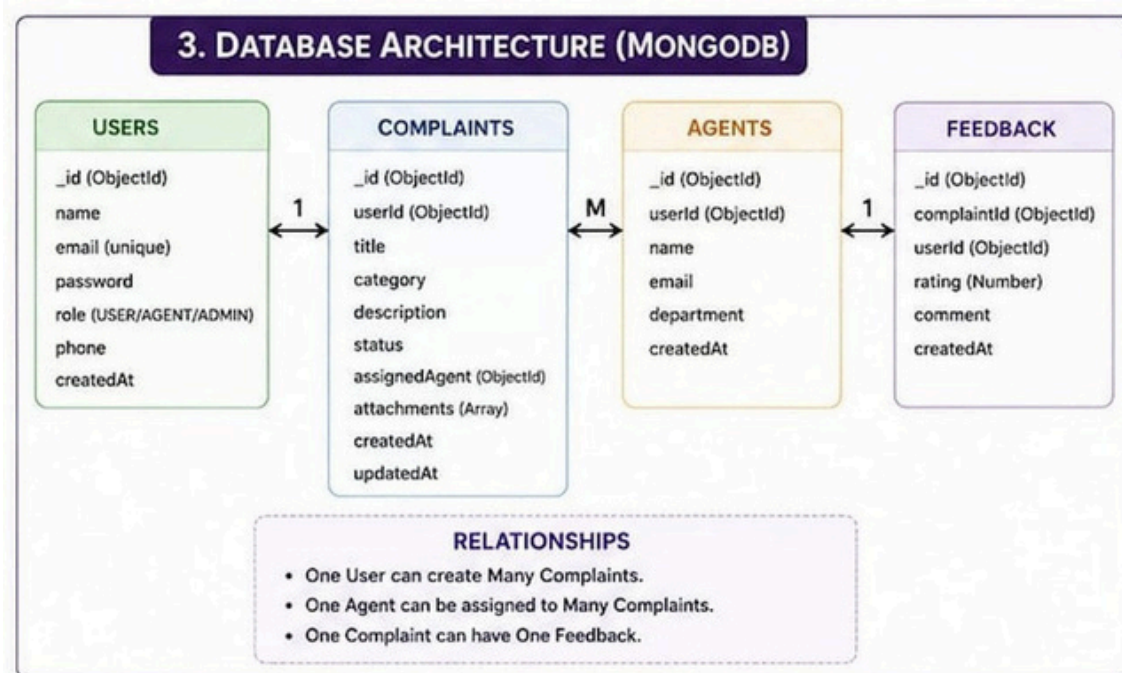
3.2 Backend Architecture

The backend follows a RESTful architecture implemented using Node.js and Express.js.

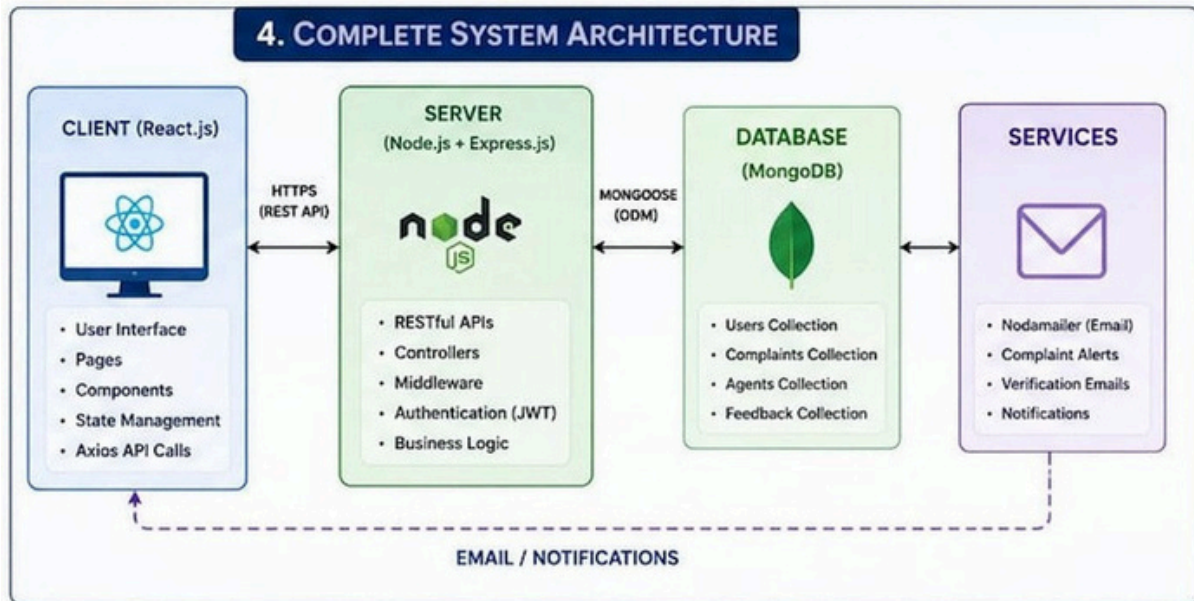


3.3 Database Architecture

MongoDB Atlas is used as the cloud-hosted database solution.



3.4 COMPLETE SYSTEM ARCHITECTURE



CHAPTER 4

SETUP AND INSTALLATION

4.1 Prerequisites

The following software and services are required:

- Node.js (v18 or higher)
- npm (v9 or higher)
- MongoDB Atlas
- Account Google OAuth
- Credentials
- Resend API Key
- Git

4.2 Installation Steps

Step 1: Clone Repository

```
git clone https://github.com/suribhavini12-cyber/Online-Complaint-Management.git
cd Online-Complaint-Management
```

Step 2: Install Dependencies

```
npm run install:all
```

Step 3: Configure Environment Variables

Create .env files for both frontend and backend and configure the required credentials.

Step 4: Run Application

```
npm run dev
```

CHAPTER 5

FOLDER STRUCTURE

Client Folder

client/

```
|—— src/
| |—— components/
| |—— context/
| |—— pages/
| |—— services/
| |—— assets/
| |—— theme.js
```

Description

- **Components** → Reusable UI components.
- **Context** → Global state management.
- **Pages** → Application screens.
- **Services** → API communication.
- **Assets** → Images and static resources.

Server Folder

server/

```
|—— controllers/
|—— middleware/
|—— models/
|—— routes/
|—— utils/
|—— server.js
```

Description

- **Controllers** → Business logic implementation.
- **Middleware** → Authentication and authorization.
- **Models** → Database schemas.
- **Routes** → API endpoint definitions.
- **Utils** → Utility and helper functions.

CHAPTER 6

RUNNING THE APPLICATION

Run Frontend and Backend Together

npm run dev

Run Backend Separately

cd server

npm run dev

Run Frontend Separately

cd client

npm run dev

CHAPTER 7

API DOCUMENTATION

Authentication APIs

Method	Endpoint	Description
POST	/api/auth/register	Register New User
POST	/api/auth/login	User Login
POST	/api/auth/google	Google Authentication

Complaint APIs

Method	Endpoint	Description
GET	/api/complaints	Retrieve Complaints
POST	/api/complaints	Create Complaint
PUT	/api/complaints/:id/status	Update Status

Feedback APIs

Method	Endpoint	Description
GET	/api/feedback/:id	Retrieve Feedback
POST	/api/feedback	Submit Feedback

CHAPTER 8

AUTHENTICATION AND AUTHORIZATION

OCRMS implements secure authentication using JSON Web Tokens (JWT).

Authentication Workflow

1. User enters credentials.
2. Server validates credentials.
3. JWT token is generated.
4. Token is stored on the client side.
5. Protected routes verify token authenticity.

User Roles

User

- Register complaints
- Track complaint progress
- Chat with agents
- Submit feedback

Agent

- View assigned complaints
- Update complaint status
- Communicate with users

Admin

- Manage users
- Monitor complaints
- Access system-wide reports

Google OAuth Integration

Google OAuth allows users to log in securely using their Google accounts.

CHAPTER 9

USER INTERFACE

Home Page

Features:

- **Animated Hero Section**
- **Feature Showcase Cards**
- **FAQ Section**
- **Call-To-Action Components**

Complaint Registration Page

Features:

- **Dynamic Form Validation**
- **Category Selection**
- **Address Details Input**

Complaint Tracking Page

Features:

- **Visual Status Stepper**
- **Progress Tracking**
- **Assigned Agent Information**

Chat System

Features:

- **Real-Time Messaging**
- **Instant Notifications**

Chatbot Assistant

Features:

- **Complaint Filing Guidance**
- **FAQ Support**
- **User Assistance**

CHAPTER 10

TESTING

Functional Testing

The following modules were tested:

- User Registration
- User Login
- Complaint Submission
- Complaint Tracking
- Chat Functionality
- Chatbot Responses
- Feedback Submission

Responsive Testing

The application was tested on:

- Desktop Devices
- Tablets
- Mobile Phones

Build Testing

Production builds were verified using:

```
npm run build
```

CHAPTER 11

SCREENSHOTS AND DEMO

Live Frontend Application

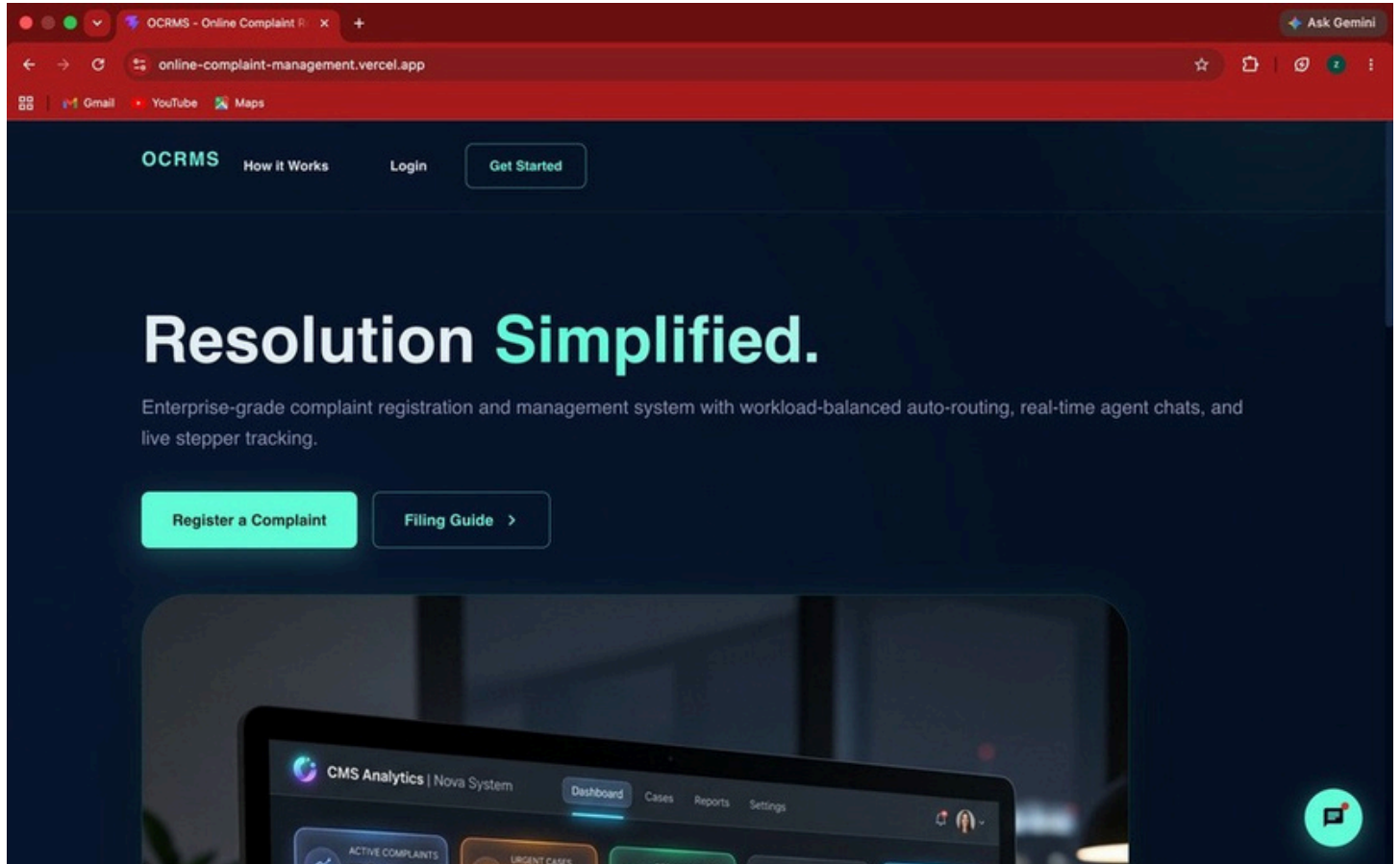
<https://online-complaint-management.vercel.app>

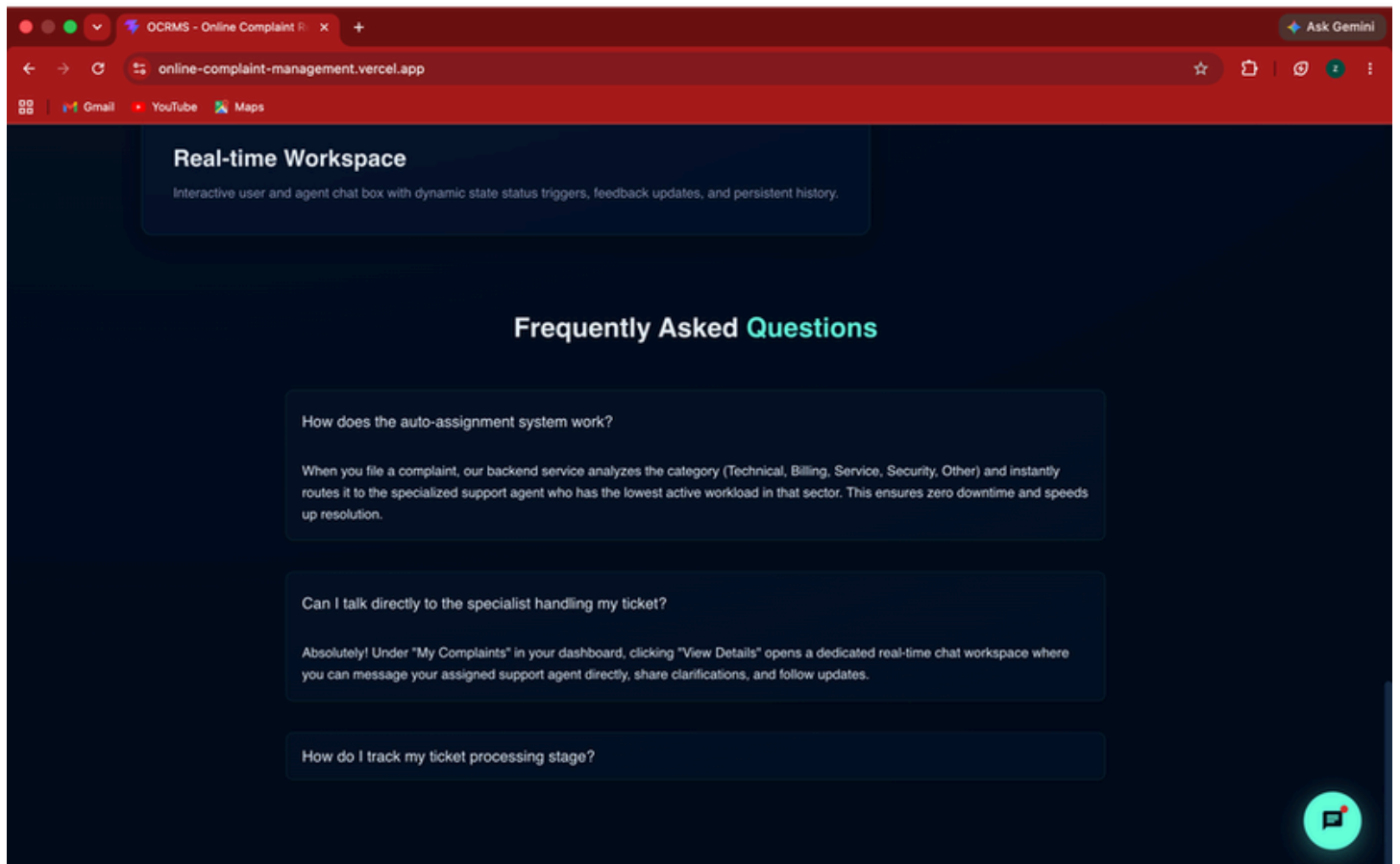
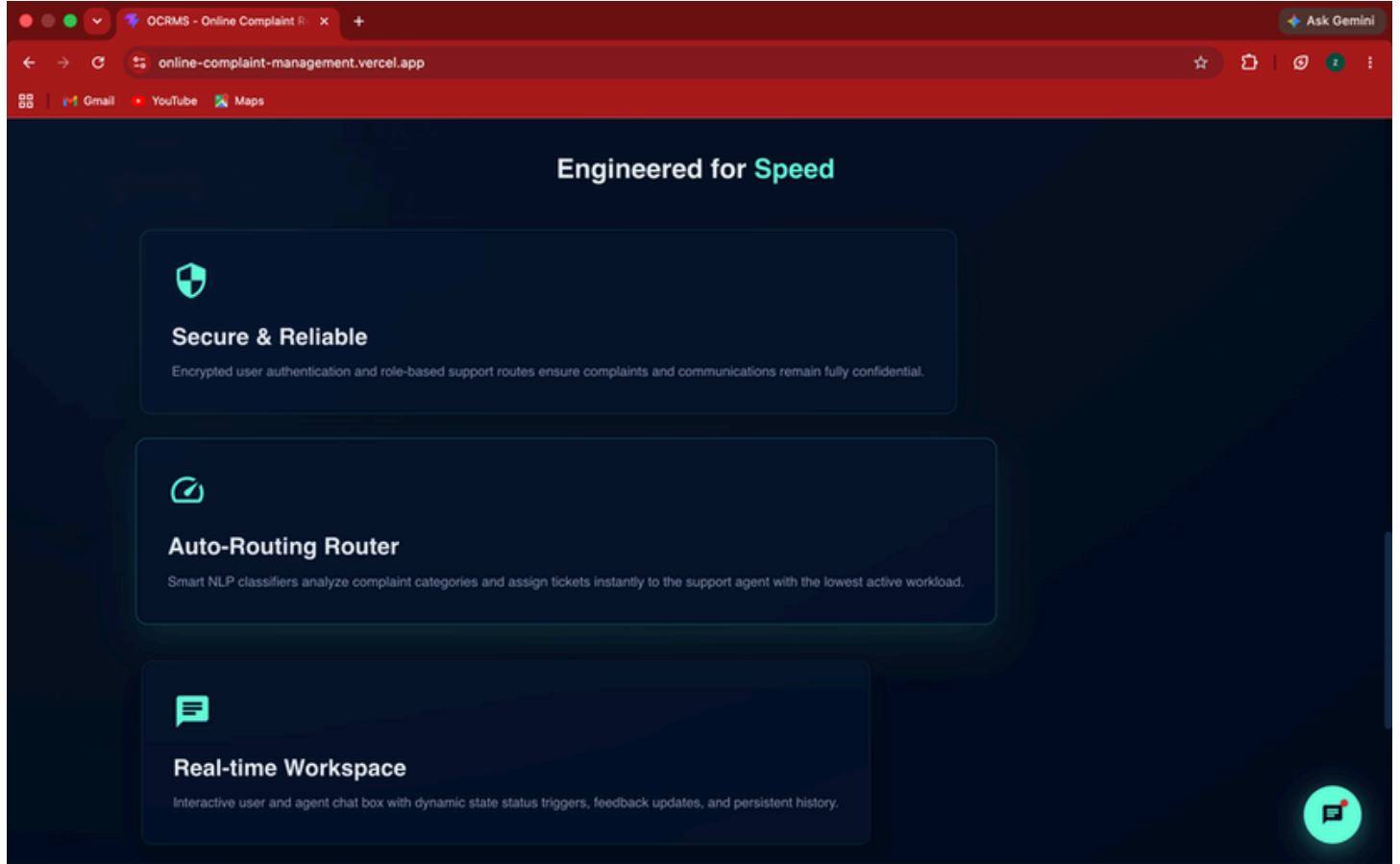
Live Backend API

<https://online-complaint-management-ggyd.onrender.com>

Screenshots to Include

1. Home Page





2. Login Page

The screenshot shows a web browser window with the URL `online-complaint-management.vercel.app/login`. The page has a dark blue background with a subtle pattern. A central white card contains the login form. At the top of the card, it says "Welcome Back" in bold, followed by the instruction "Enter your credentials to access your account". Below this are two input fields: "Email Address *" and "Password *". A red "Login" button is positioned below the password field. Underneath the button is the text "OR". Below "OR" is a "Sign in as zero" button with a profile picture and the email `zorokutiy85003@gmail.com`. At the bottom of the card, there are two links: "Don't have an account? Register here" and "Forgot Password?". A small red chat icon is visible in the bottom right corner of the page.

OCRMS - Online Complaint Resolution Management System

online-complaint-management.vercel.app/login

Ask Gemini

Google Gmail YouTube Maps

Welcome Back

Enter your credentials to access your account

Email Address *

Password *

Login

OR

Sign in as zero
zorokutiy85003@gmail.com

Don't have an account? Register here

Forgot Password?

3.Registration Page

The screenshot shows a web browser window with the URL `online-complaint-management.vercel.app/register`. The page has a dark blue background with a subtle pattern. A central white card contains the registration form. At the top of the card, it says "Create Account" in bold, followed by the instruction "Join the OCRMS resolution network". Below this are four input fields: "Full Name *", "Email Address *", "Phone Number *", and "Password *". Below the password field is a "Join As" section with a dropdown menu currently showing "Individual User". A red "Register" button is positioned below the "Join As" section. At the bottom of the card, there is a link: "Already have an account? Login here". A small red chat icon is visible in the bottom right corner of the page.

OCRMS - Online Complaint Resolution Management System

online-complaint-management.vercel.app/register

Ask Gemini

Google Gmail YouTube Maps

Create Account

Join the OCRMS resolution network

Full Name *

Email Address *

Phone Number *

Password *

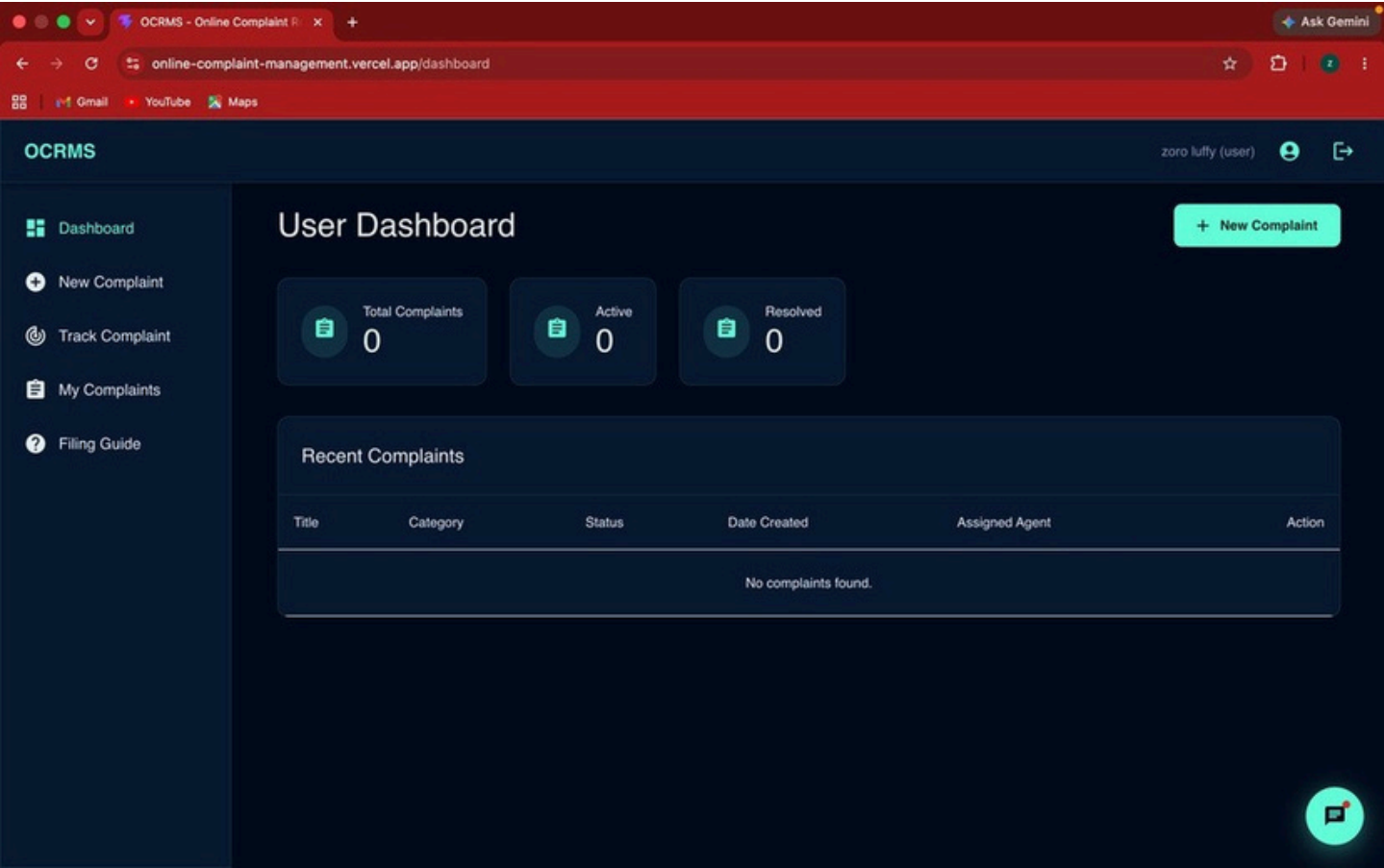
Join As

Individual User

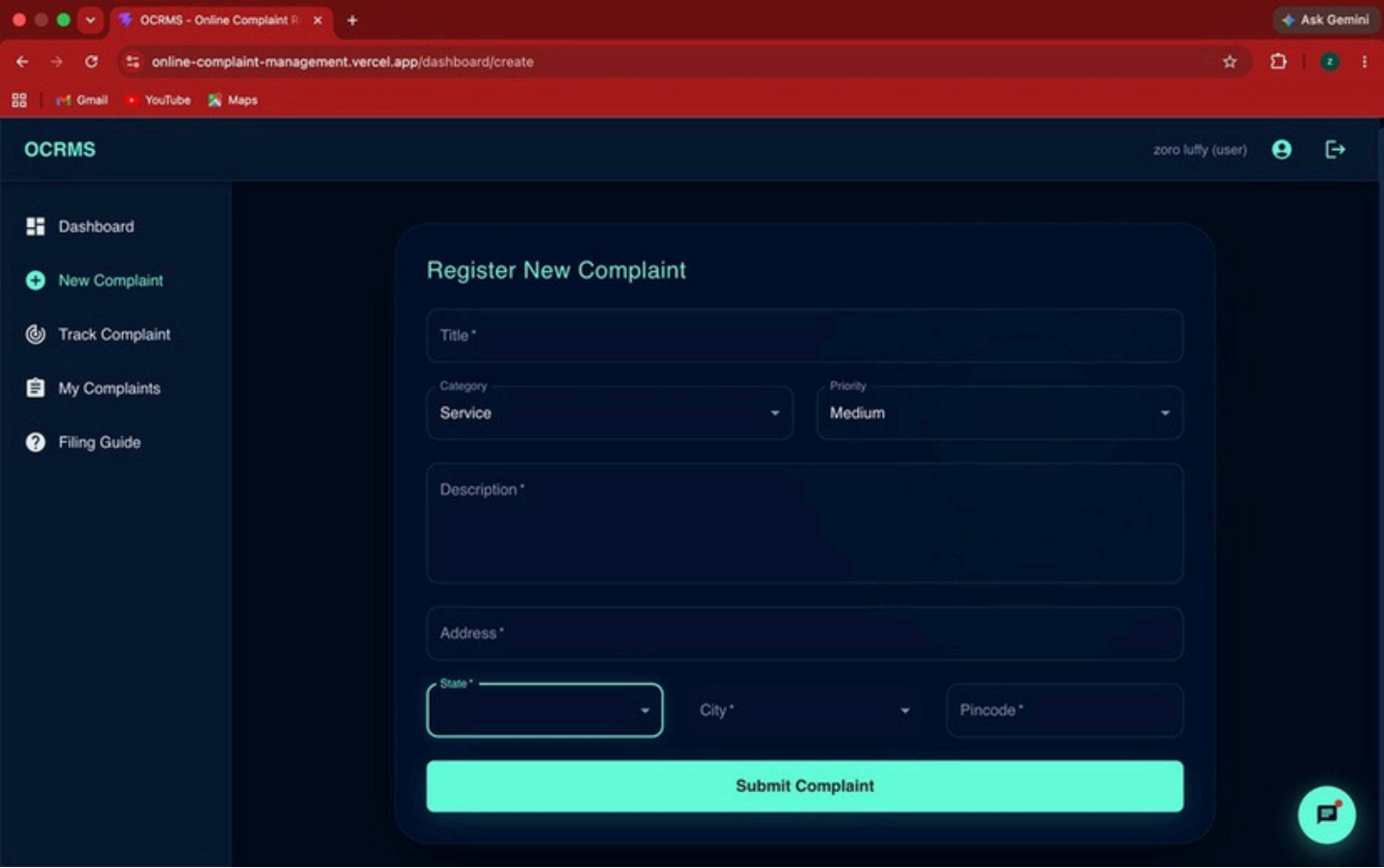
Register

Already have an account? Login here

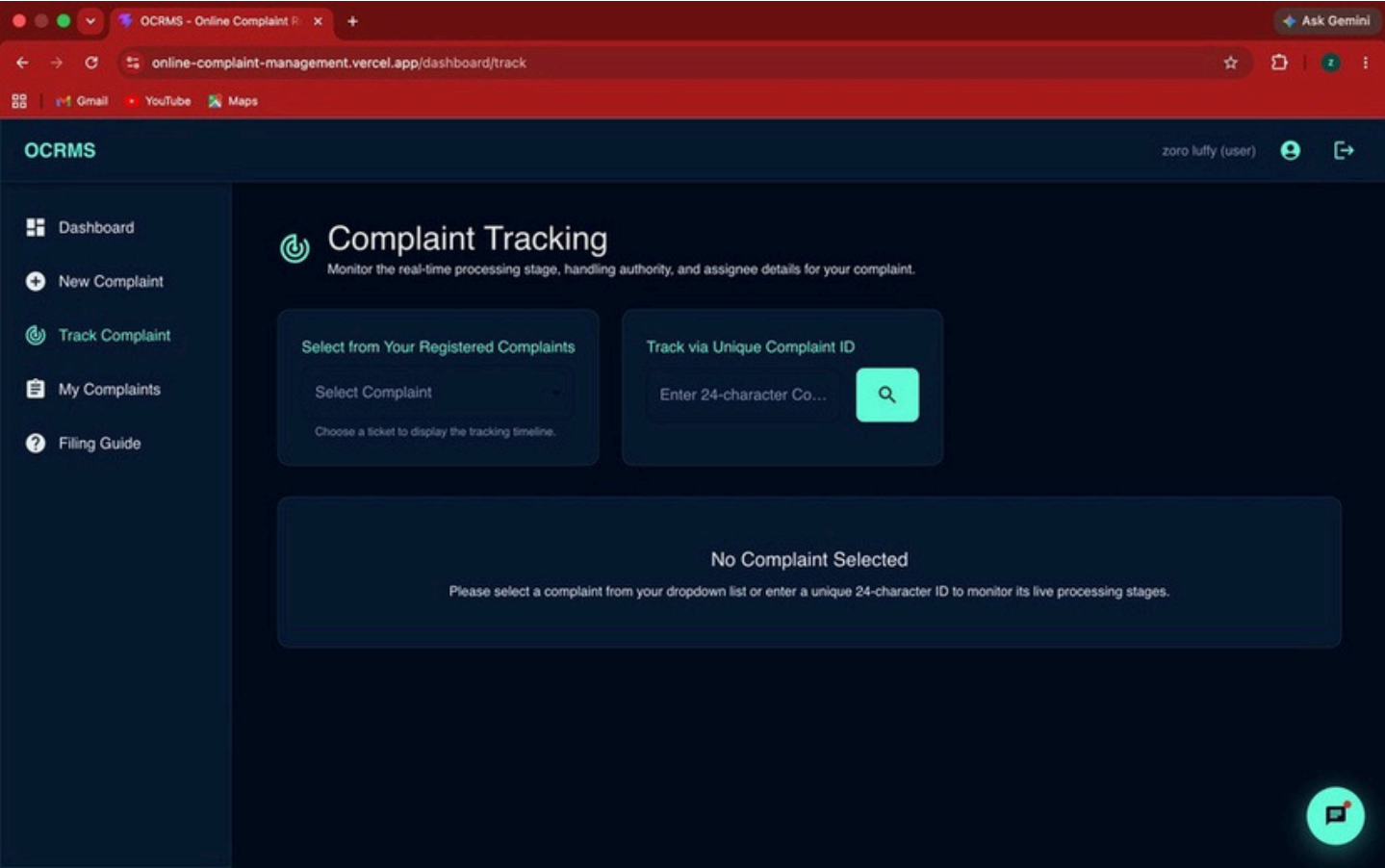
4.User Dashboard



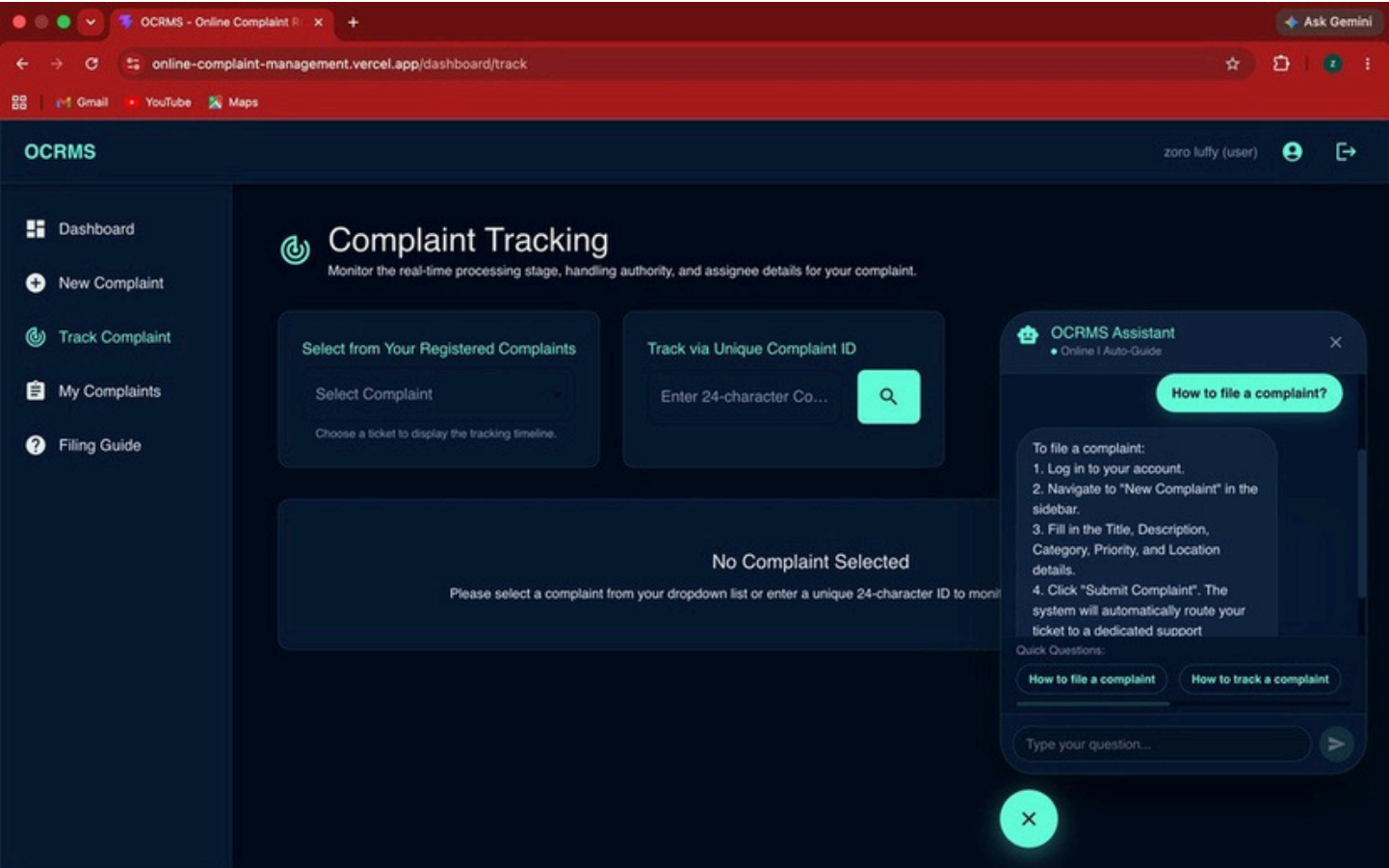
5.Complaint Registration Page



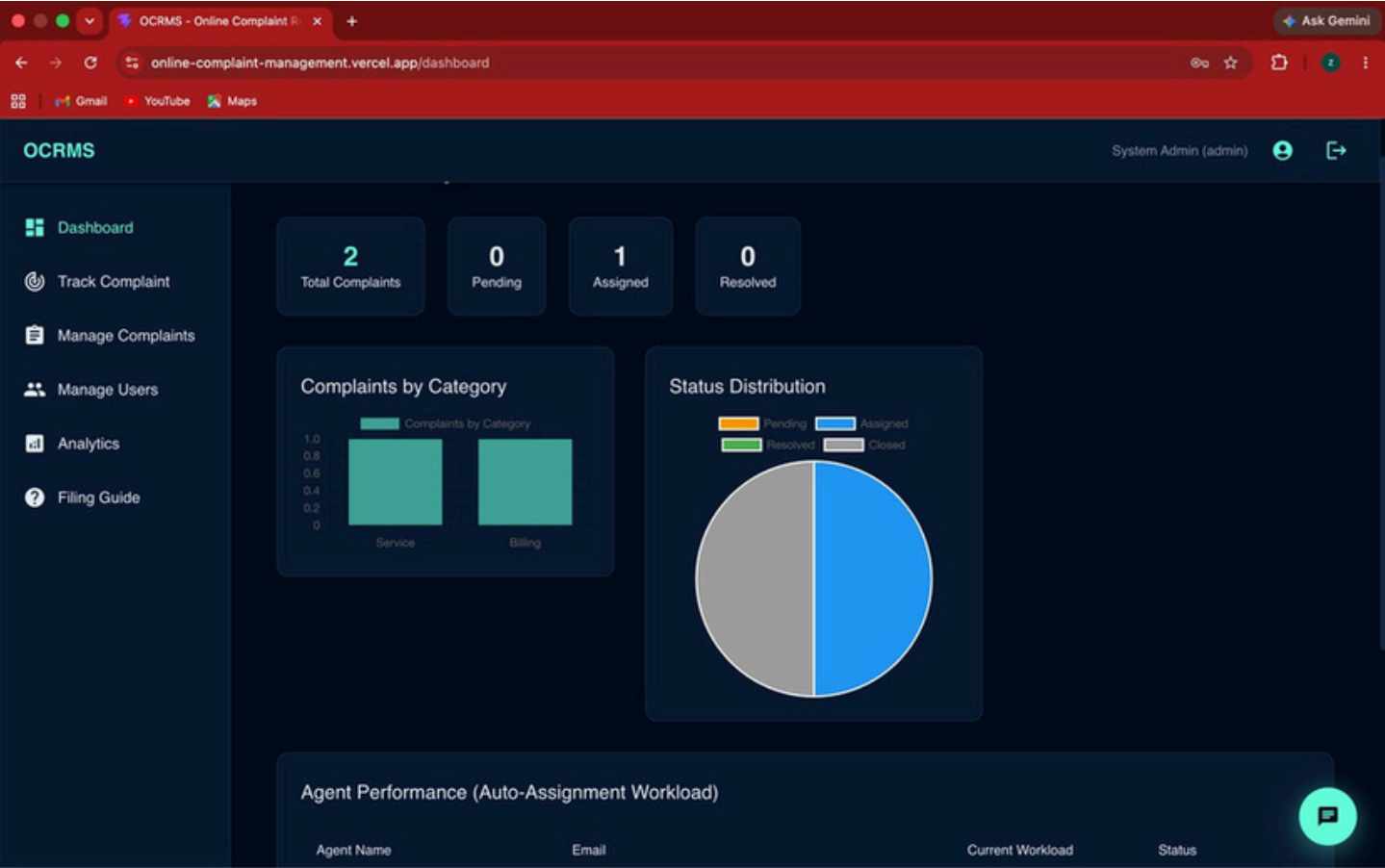
6.Complaint Tracking Page



7.Real-Time Chat Interface & ChatBot



8.Admin Dashboard



The screenshot shows the OCRMS Admin Dashboard with the 'Manage Users' section selected. The top navigation bar includes the OCRMS logo, the user 'System Admin (admin)', and a 'Ask Gemini' button. The left sidebar contains links to Dashboard, Track Complaint, Manage Complaints, Manage Users, Analytics, and Filing Guide. The main content area features a search bar with the text 'Search by Name, Em...' and a 'Filter by Role' dropdown menu set to 'All Roles'. Below the search bar is a table listing users with columns for Name, Email, Phone, Role, Workload (Agents Only), and Actions.

Name	Email	Phone	Role	Workload (Agents Only)	Actions
System Admin	admin@ocrms.com	1234567890	ADMIN	-	 
Billing Specialist	billing_agent@ocrms.com	1234567892	AGENT	1 Low	 
Technical Specialist	tech_agent@ocrms.com	1234567891	AGENT	0 Low	 
Service Specialist	service_agent@ocrms.com	1234567894	AGENT	1 Low	 
Security Specialist	security_agent@ocrms.com	1234567895	AGENT	0 Low	 
General Specialist	general_agent@ocrms.com	1234567896	AGENT	0 Low	 
John Doe	user@ocrms.com	1234567893	USER	-	 

OCRMS - Online Complaint R...x

online-complaint-management.vercel.app/dashboard/complaints

Ask Gemini

GmailYouTubeMaps

OCRMS

System Admin (admin)

Dashboard

Track Complaint

Manage Complaints

Manage Users

Analytics

Filing Guide

Manage Complaints

Search Title, Descript...

Status

All Statuses

Category

All Categories

Priority

All Priorities

Complaint ID	Customer	Title	Category	Priority	Status	Date Created	Assigned Agent	Action
#6A01C2	John Doe	Service	Service	Medium	Assigned	18/06/2026	Service Specialist	
#EF2D14	Komal Sushank	bill	Billing	Medium	Closed	18/06/2026	Billing Specialist	

OCRMS - Online Complaint R...x

online-complaint-management.vercel.app/dashboard/analytics

Ask Gemini

GmailYouTubeMaps

OCRMS

System Admin (admin)

Dashboard

Track Complaint

Manage Complaints

Manage Users

Analytics

Filing Guide

System Analytics

2

Total Complaints

0

Pending

1

Assigned

0

Resolved

Complaints by Category

Complaints by Category

Category	Count
Billing	1
Service	1

Status Distribution

PendingAssignedResolvedClosed

Status	Count
Pending	1
Assigned	1
Resolved	1
Closed	1

CHAPTER 12

KNOWN ISSUES

HostingDelay

The application is deployed using free-tier cloud services. Due to inactivity policies:

- **Render services may enter sleep mode.**
- **MongoDB Atlas free clusters may experience cold starts.**

This may cause the first request after inactivity to take additional time.

CHAPTER 13

FUTURE ENHANCEMENTS

Future improvements planned for OCRMS include:

- **AI-Based Complaint Classification**
- **Multi-Language Support**
- **Mobile Application Development**
- **Push Notifications**
- **Advanced Analytics Dashboard**
- **Complaint Escalation Workflow**
- **Sentiment Analysis for Complaints**
- **AI-Powered Resolution Suggestions**
- **Voice-Based Complaint Registration**
- **Integration with Government Service Portals**

CHAPTER 14

DEPLOYMENT

Frontend Deployment

Platform: Vercel

Backend Deployment

Platform: Render

Database Hosting

Platform: MongoDBAtlas

The deployment architecture ensures scalability, availability, and secure cloud-based access for users.

CHAPTER 15

CONCLUSION

TheOnlineComplaintRegistration and Management System (OCRMS) successfully provides a modern and efficient platform for complaint handling. Through automated complaint assignment, real-time communication, secure authentication, and transparent tracking mechanisms, the system enhances customer satisfaction and operational efficiency. modern and efficient platform for complaint handling. Through automated complaint assignment, real-time communication, secure authentication, and transparent tracking mechanisms, the system enhances customer satisfaction and operational efficiency.